

INTERNATIONAL STANDARD

IEC
60384-6-1

QC 300501

Second edition
2005-09

Fixed capacitors for use in electronic equipment –

Part 6-1:

Blank detail specification:

Fixed metallized polycarbonate film dielectric d.c. capacitors –

Assessment level E



Reference number
IEC 60384-6-1:2005(E)

Publication numbering

As from 1 January 1997 all IEC publications are issued with a designation in the 60000 series. For example, IEC 34-1 is now referred to as IEC 60034-1.

Consolidated editions

The IEC is now publishing consolidated versions of its publications. For example, edition numbers 1.0, 1.1 and 1.2 refer, respectively, to the base publication, the base publication incorporating amendment 1 and the base publication incorporating amendments 1 and 2.

Further information on IEC publications

The technical content of IEC publications is kept under constant review by the IEC, thus ensuring that the content reflects current technology. Information relating to this publication, including its validity, is available in the IEC Catalogue of publications (see below) in addition to new editions, amendments and corrigenda. Information on the subjects under consideration and work in progress undertaken by the technical committee which has prepared this publication, as well as the list of publications issued, is also available from the following:

- **IEC Web Site** (www.iec.ch)

- **Catalogue of IEC publications**

The on-line catalogue on the IEC web site (www.iec.ch/searchpub) enables you to search by a variety of criteria including text searches, technical committees and date of publication. On-line information is also available on recently issued publications, withdrawn and replaced publications, as well as corrigenda.

- **IEC Just Published**

This summary of recently issued publications (www.iec.ch/online_news/justpub) is also available by email. Please contact the Customer Service Centre (see below) for further information.

- **Customer Service Centre**

If you have any questions regarding this publication or need further assistance, please contact the Customer Service Centre:

Email: custserv@iec.ch
Tel: +41 22 919 02 11
Fax: +41 22 919 03 00

INTERNATIONAL STANDARD

IEC 60384-6-1

QC 300501

Second edition
2005-09

Fixed capacitors for use in electronic equipment –

Part 6-1:

Blank detail specification:

Fixed metallized polycarbonate film dielectric d.c. capacitors –

Assessment level E

© IEC 2005 — Copyright - all rights reserved

No part of this publication may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying and microfilm, without permission in writing from the publisher.

International Electrotechnical Commission, 3, rue de Varembé, PO Box 131, CH-1211 Geneva 20, Switzerland
Telephone: +41 22 919 02 11 Telefax: +41 22 919 03 00 E-mail: inmail@iec.ch Web: www.iec.ch



Commission Electrotechnique Internationale
International Electrotechnical Commission
Международная Электротехническая Комиссия

PRICE CODE

M

For price, see current catalogue

INTERNATIONAL ELECTROTECHNICAL COMMISSION

FIXED CAPACITORS FOR USE IN ELECTRONIC EQUIPMENT –**Part 6-1: Blank detail specification:
Fixed metallized polycarbonate film dielectric d.c. capacitors –
Assessment level E**

FOREWORD

- 1) The International Electrotechnical Commission (IEC) is a worldwide organization for standardization comprising all national electrotechnical committees (IEC National Committees). The object of IEC is to promote international co-operation on all questions concerning standardization in the electrical and electronic fields. To this end and in addition to other activities, IEC publishes International Standards, Technical Specifications, Technical Reports, Publicly Available Specifications (PAS) and Guides (hereafter referred to as "IEC Publication(s)"). Their preparation is entrusted to technical committees; any IEC National Committee interested in the subject dealt with may participate in this preparatory work. International, governmental and non-governmental organizations liaising with the IEC also participate in this preparation. IEC collaborates closely with the International Organization for Standardization (ISO) in accordance with conditions determined by agreement between the two organizations.
- 2) The formal decisions or agreements of IEC on technical matters express, as nearly as possible, an international consensus of opinion on the relevant subjects since each technical committee has representation from all interested IEC National Committees.
- 3) IEC Publications have the form of recommendations for international use and are accepted by IEC National Committees in that sense. While all reasonable efforts are made to ensure that the technical content of IEC Publications is accurate, IEC cannot be held responsible for the way in which they are used or for any misinterpretation by any end user.
- 4) In order to promote international uniformity, IEC National Committees undertake to apply IEC Publications transparently to the maximum extent possible in their national and regional publications. Any divergence between any IEC Publication and the corresponding national or regional publication shall be clearly indicated in the latter.
- 5) IEC provides no marking procedure to indicate its approval and cannot be rendered responsible for any equipment declared to be in conformity with an IEC Publication.
- 6) All users should ensure that they have the latest edition of this publication.
- 7) No liability shall attach to IEC or its directors, employees, servants or agents including individual experts and members of its technical committees and IEC National Committees for any personal injury, property damage or other damage of any nature whatsoever, whether direct or indirect, or for costs (including legal fees) and expenses arising out of the publication, use of, or reliance upon, this IEC Publication or any other IEC Publications.
- 8) Attention is drawn to the Normative references cited in this publication. Use of the referenced publications is indispensable for the correct application of this publication.
- 9) Attention is drawn to the possibility that some of the elements of this IEC Publication may be the subject of patent rights. IEC shall not be held responsible for identifying any or all such patent rights.

International Standard IEC 60384-6-1 has been prepared by IEC technical committee 40: Capacitors and resistors for electronic equipment.

This second edition of IEC 60384-6-1 cancels and replaces the first edition published in 1987. It constitutes a technical revision. All changes that have been agreed upon can be categorized as minor revisions related to tables, figures and references..

The text of this standard is based on the following documents:

FDIS	Report on voting
40/1589/FDIS	40/1609/RVD

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

The QC number that appears on the front cover of this publication is the specification number in the IEC Quality Assessment System for Electronic Components (IECQ).

This publication should be read in conjunction with IEC 60384-1 and IEC 60384-6.

This standard forms Part 6-1 of IEC 60384 which is published under the general title *Fixed capacitors for use in electronic equipment*.

Part 6 is composed as follows:

Part 6: Sectional specification – Fixed metallized polycarbonate film dielectric d.c. capacitors

Part 6-1: Blank detail specification – Fixed metallized polycarbonate film dielectric d.c. capacitors – Assessment level E

NOTE For the complete list of parts, see IEC 60384-6.

The committee has decided that the contents of this publication will remain unchanged until the maintenance result date indicated on the IEC web site under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

- reconfirmed;
- withdrawn;
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

FIXED CAPACITORS FOR USE IN ELECTRONIC EQUIPMENT –

Part 6-1: Blank detail specification: Fixed metallized polycarbonate film dielectric d.c. capacitors – Assessment level E

INTRODUCTION

Blank detail specification

A blank detail specification is a supplementary document to the sectional specification and contains requirements for style and layout and minimum content of detail specifications. Detail specifications not complying with these requirements may not be considered as being in accordance with IEC specifications nor shall they so be described.

In the preparation of detail specifications, the content of 1.3 of the sectional specification shall be taken into account. The numbers between brackets on the first page correspond to the following information which shall be inserted in the position indicated.

Identification of the detail specification

- [1] The International Electrotechnical Commission or the National Standards Organization under whose authority the detail specification is published.
- [2] The IEC or National Standards number of the detail specification, date of issue and any further information required by the national system.
- [3] The number and issue number of the IEC or national generic specification.
- [4] The IEC number of the blank detail specification.

Identification of the capacitor

- [5] A short description of the type of capacitor.
- [6] Information on typical construction (when applicable).

NOTE When the capacitor is not designed for use in printed board applications, this should be clearly stated in the detail specification in this position.

- [7] Outline drawing with main dimensions which are of importance for interchangeability and/or reference to the national or international documents for outlines. Alternatively, this drawing may be given in an appendix to the detail specification.
- [8] Application or group of applications covered and/or assessment level.

NOTE The assessment level(s) to be used in a detail specification is/are selected from 3.5.4 of the sectional specification. This implies that one blank detail specification may be used in combination with several assessment levels, provided the grouping of the tests does not change.

- [9] Reference data on the most important properties, to allow comparison between the various capacitor types.

[1]	IEC 60384-6-1-XXX [2] QC 300501-XXX
ELECTRONIC COMPONENTS OF ASSESSED QUALITY IN ACCORDANCE WITH: [3]	IEC 60384-6-1 [4] QC 300501
	FIXED METALLIZED [5] POLYCARBONATE FILM DIELECTRIC DC CAPACITORS
Outline drawing: (see Table 1) (...angle projection) [7] (Other shapes are permitted within the dimensions given) For notes [], see preceding page.	[6]
	Assessment level(s): E [8] Performance grade:

REFERENCE DATA:

[9]

Information on the availability of components qualified to this detail specification is given in IEC QC 001005
--

1 General data

1.1 Recommended method(s) of mounting

See 1.3.1.1 of IEC 60384-6.

1.2 Dimensions

Table 1 – Case-size reference and dimensions

Case size reference	Dimensions						
	mm						
	$\varnothing$	<i>L</i>	<i>H</i>	<i>d</i>			

NOTE 1 When there is no case-size reference, Table 1 may be omitted and the dimensions should be given in Table 2, which then becomes Table 1.

NOTE 2 The dimensions should be given as maximum dimensions or as nominal dimensions with the tolerance.

1.3 Ratings and characteristics

Capacitance range (see Table 2)

Tolerance on rated capacitance

Rated voltage (see Table 2)

Category voltage (if applicable) (see Table 2)

Climatic category

Rated temperature

Maximum a.c. voltage (if applicable)

Maximum pulse load

Tangent of loss angle

Insulation resistance

Table 2 – Values of capacitance and of voltage related to case sizes

Rated voltage				
Category voltage ^a				
Rated capacitance (in nF and/or µF)	Case size	Case size	Case size	Case size

^a If different from the rated voltage.

1.4 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60384-1: *Fixed capacitors for use in electronic equipment – Part 1: Generic specification*

IEC 60384-6: *Fixed capacitors for use in electronic equipment – Part 6: Sectional specification – Fixed metallized polycarbonate film dielectric d.c. capacitors*

IEC 60410 (all parts), *Terms and nomenclature for cores made of magnetically soft ferrites*

1.5 Marking

The marking of the capacitor and the package shall be in accordance with the requirements of 1.5 of IEC 60384-6.

NOTE The details of the marking of the component and package should be given in full in the detail specification.

1.6 Ordering information

Orders for capacitors covered by this specification shall contain, in clear or in coded form, the following minimum information:

- a) rated capacitance;
- b) tolerance on rated capacitance;
- c) rated d.c. voltage;
- d) number and issue reference of the detail specification and style reference.

1.7 Certified records of released lots

Required/not required.

1.8 Additional information

(not for inspection purposes)

1.9 Additional or increased severities or requirements to those specified in the generic and/or sectional specification

NOTE Additions or increased requirements should be specified only when essential.

Table 3 – Other characteristics

This table is to be used for defining characteristics which are additional to, or more severe than, those given in the sectional specification.

2 Inspection requirements

2.1 Procedures

2.1.1 For qualification approval, the procedures shall be in accordance with 3.4 of IEC 60384-6.

2.1.2 For quality conformance inspection, the test schedule (Table 4) includes sampling, periodicity, severities and requirements. The formation of inspection lots is covered by 3.5.1 of IEC 60384-6.

Table 4 – Test schedule for quality conformance inspection

Subclause number and test ^a	D or ND	Conditions of test ^a	IL b	AQL	Performance requirements ^a
Group A inspection (lot-by-lot) Subgroup A1 4.1 Visual examination 4.1 Dimensions (gauging)	ND		S-4	2,5 %	As in 4.1 Legible marking and as specified in 1.5 of this specification As specified in Table 1 of this specification
Subgroup A2 4.2.2 Capacitance 4.2.3 Tangent of loss angle 4.2.1 Voltage proof (Test A) 4.2.4 Insulation resistance (Test A)	ND		II	1,0 %	Within specified tolerance As in 4.2.3.2 No breakdown or flashover As in 4.2.4.2
Group B inspection (lot-by-lot) Subgroup B1 4.5 Solderability 4.15 Solvent resistance of the marking (if applicable)	ND	Without ageing Method: ... Solvent: ... Solvent temperature: ... Method 1 Rubbing material: ... Recovery time:	S-3	2,5	Good tinning as evidenced by free flowing of the solder with wetting of the terminations or solder shall flow within ... s, as applicable Legible marking

Subclause number and test ^a	D or ND	Conditions of test ^a	Sample size and criterion of acceptability ^c			Performance requirements ^a
			<i>p</i>	<i>n</i>	<i>c</i>	
Group C inspection (periodic)						
Subgroup C1A Part of sample of subgroup C1	D		6	9	1	
4.1 Dimensions (detail)						See detail specification
4.3.1 Initial measurements		Capacitance Tangent of loss angle: For $C_R > 1 \mu\text{F}$: at 1 kHz $C_R \leq 1 \mu\text{F}$: at 10 kHz				
4.3 Robustness of terminations		Visual examination				No visible damage
4.4 Resistance to soldering heat		Method:				
4.14 Component solvent resistance (if applicable)		Solvent: ... Solvent temperature: ... Method 2 Recovery time:				See detail specification
4.4.2 Final measurements		Visual examination				No visible damage Legible marking
		Capacitance				$\Delta C/C \leq 1\%$ for Grade 1 $\leq 2\%$ for Grade 2 of value measured initially
		Tangent of loss angle				Increase of $\tan \delta$: $\leq 0,003$ $C \leq 1 \mu\text{F}$ Grade 1 $\leq 0,002$ $C > 1 \mu\text{F}$ Grade 1 $\leq 0,005$ $C \leq 1 \mu\text{F}$ Grade 2 $\leq 0,003$ $C > 1 \mu\text{F}$ Grade 2 compared to values measured in 4.3.1
Subgroup C1B Other part of sample of subgroup C1	D		6	18	1	
4.6.1 Initial measurements		Capacitance Tangent of loss angle: For $C_R > 1 \mu\text{F}$: at 1 kHz $C_R \leq 1 \mu\text{F}$: at 10 kHz				
4.6 Rapid change of temperature		T_A = Lower category temperature T_B = Upper category temperature Five cycles Duration $t = 30$ min Visual examination				
4.7 Vibration		Method of mounting: see 1.1 of this specification Frequency range: ... Hz to ... Hz Amplitude 0,75 mm or acceleration 100 m/s^2 (whichever is the less severe) Total duration: 6 h				No visible damage

Subclause number and test ^a	D or ND	Conditions of test ^a	Sample size and criterion of acceptability ^c			Performance requirements ^a
			<i>p</i>	<i>n</i>	<i>c</i>	
4.7.2 Final inspection		Visual examination				No visible damage
4.8 Bump (or shock, see 4.9)		Method of mounting: see 1.1 of this specification				
		Number of bumps: ... Acceleration: ... m/s ² Duration of pulse: ... ms				
4.9 Shock (or bump, see 4.8)		Method of mounting: see 1.1 of this specification				
		Acceleration: ... m/s ² Duration of pulse: ... ms				
4.8.3 Final or 4.9.3 measurements		Visual examination				No visible damage
		Capacitance				$\Delta C/C \leq 1\%$ for Grade 1 $\leq 2\%$ for Grade 2 of value measured initially
		Tangent of loss angle				Increase of $\tan \delta$: $\leq 0,003$ $C \leq 1 \mu F$ Grade 1 $\leq 0,002$ $C > 1 \mu F$ Grade 1 $\leq 0,005$ $C \leq 1 \mu F$ Grade 2 $\leq 0,003$ $C > 1 \mu F$ Grade 2 compared to values measured in 4.6.1
		Insulation resistance				$\geq 50\%$ of values in 4.2.4.2
Subgroup C1 Combined sample of specimens of subgroups C1A and C1B	D		6	27	2	
4.10 Climatic sequence						
4.10.2 Dry heat		Temperature: upper category temperature Duration: 16 h				
4.10.3 Damp heat, cyclic, Test Db, first cycle						
4.10.4 Cold		Temperature: lower category temperature Duration: 2 h				
4.10.5 Low air pressure (if required by the detail specification)		Air pressure: 8 kPa				
4.10.5.3 Final measurement		Visual examination				No permanent breakdown, flashover or harmful deformation of the case
4.10.6 Damp heat, cyclic, Test dB, remaining cycles						

Subclause number and test ^a	D or ND	Conditions of test ^a	Sample size and criterion of acceptability ^c			Performance requirements ^a
			<i>p</i>	<i>n</i>	<i>c</i>	
4.10.6.1 Final measurements		Visual examination Capacitance Tangent of loss angle Insulation resistance				No visible damage Legible marking $\Delta C/C \leq 3\%$ for Grade 1 $\leq 5\%$ for Grade 2 of value measured in 4.4.2, 4.8.3 or 4.9.3 as applicable Increase of $\tan \delta$: $\leq 0,005$ $C \leq 1 \mu\text{F}$ Grade 1 $\leq 0,003$ $C > 1 \mu\text{F}$ Grade 1 $\leq 0,008$ $C \leq 1 \mu\text{F}$ Grade 2 $\leq 0,005$ $C > 1 \mu\text{F}$ Grade 2 compared to values measured in 4.3.1 or 4.6.1 as applicable $\geq 50\%$ of values in 4.2.4.2
Subgroup C2 4.11 Damp heat, steady state 4.11.1 Initial measurements 4.11.3 Final measurements	D	Capacitance Tangent of loss angle at 1 kHz Visual examination Capacitance Tangent of loss angle Insulation resistance	6	15	1	No visible damage Legible marking $\Delta C/C \leq 3\%$ for Grade 1 $\leq 5\%$ for Grade 2 of value measured in 4.11.1 Increase of $\tan \delta$: $\leq 0,005$ compared to values measured in 4.11.1 $\geq 50\%$ of values in 4.2.4.2
Subgroup C3 4.12 Endurance 4.12.1 Initial measurements 4.12.5 Final measurements	D	Duration: Grade 1: 2 000 h Grade 2: 1 000 h Capacitance Tangent of loss angle: For $C_R > 1 \mu\text{F}$: at 1 kHz $C_R \leq 1 \mu\text{F}$: at 10 kHz Visual examination Capacitance Tangent of loss angle Insulation resistance	3	21	1	No visible damage Legible marking $\Delta C/C \leq 3\%$ for Grade 1 $\leq 5\%$ for Grade 2 of value measured in 4.12.1 Increase of $\tan \delta$: $\leq 0,003$ $C \leq 1 \mu\text{F}$ Grade 1 $\leq 0,002$ $C > 1 \mu\text{F}$ Grade 1 $\leq 0,005$ $C \leq 1 \mu\text{F}$ Grade 2 $\leq 0,003$ $C > 1 \mu\text{F}$ Grade 2 compared to values measured in 4.12.1 $\geq 50\%$ of values in 4.2.4.2

Subclause number and test ^a	D or ND	Conditions of test ^a	Sample size and criterion of acceptability ^c			Performance requirements ^a
			<i>p</i>	<i>n</i>	<i>c</i>	
Subgroup C4	D		6	15	1	
4.2.5 Characteristics depending on temperature (if applicable)		Capacitance insulation resistance (Grade 1 only)				As in 4.2.5
4.13 Charge and discharge						
4.13.1 Initial measurements		Capacitance Tangent of loss angle: For $C_R > 1 \mu\text{F}$: at 1 kHz $C_R \leq 1 \mu\text{F}$: at 10 kHz Duration of charge: ... s Duration of discharge: ... s				
4.13.3 Final measurements		Capacitance Tangent of loss angle Insulation resistance				$\Delta C/C \leq 2 \%$ for Grade 1 $\leq 3 \%$ for Grade 2 of value measured in 4.13.1 Increase of $\tan \delta$: $\leq 0,003$ $C \leq 1 \mu\text{F}$ Grade 1 $\leq 0,002$ $C > 1 \mu\text{F}$ Grade 1 $\leq 0,005$ $C \leq 1 \mu\text{F}$ Grade 2 $\leq 0,003$ $C > 1 \mu\text{F}$ Grade 2 compared to values measured in 4.13.1 $\geq 50 \%$ of values in 4.2.4.2
^a Subclause numbers of tests and performance requirements refer to IEC 60384-6 and Clause 1 of this specification. ^b <i>ILs</i> and <i>AQLs</i> are selected from IEC 60410. ^c In this table: <i>p</i> = periodicity (in months) <i>n</i> = sample size <i>c</i> = acceptance criterion (permitted number of nonconformances) D = destructive ND = non-destructive <i>IL</i> = inspection level <i>AQL</i> = acceptable quality level						



Standards Survey

The IEC would like to offer you the best quality standards possible. To make sure that we continue to meet your needs, your feedback is essential. Would you please take a minute to answer the questions overleaf and fax them to us at +41 22 919 03 00 or mail them to the address below. Thank you!

Customer Service Centre (CSC)

International Electrotechnical Commission

3, rue de Varembe
1211 Genève 20
Switzerland

or

Fax to: **IEC/CSC** at +41 22 919 03 00

Thank you for your contribution to the standards-making process.

A Prioritaire

Nicht frankieren
Ne pas affranchir



Non affrancare
No stamp required

RÉPONSE PAYÉE

SUISSE

Customer Service Centre (CSC)
International Electrotechnical Commission
3, rue de Varembe
1211 GENEVA 20
Switzerland



Q1 Please report on **ONE STANDARD** and **ONE STANDARD ONLY**. Enter the exact number of the standard: (e.g. 60601-1-1)

.....

Q2 Please tell us in what capacity(ies) you bought the standard (tick all that apply). I am the/a:

- purchasing agent ☐
 librarian ☐
 researcher ☐
 design engineer ☐
 safety engineer ☐
 testing engineer ☐
 marketing specialist ☐
 other

Q3 I work for/in/as a:
(tick all that apply)

- manufacturing ☐
 consultant ☐
 government ☐
 test/certification facility ☐
 public utility ☐
 education ☐
 military ☐
 other

Q4 This standard will be used for:
(tick all that apply)

- general reference ☐
 product research ☐
 product design/development ☐
 specifications ☐
 tenders ☐
 quality assessment ☐
 certification ☐
 technical documentation ☐
 thesis ☐
 manufacturing ☐
 other

Q5 This standard meets my needs:
(tick one)

- not at all ☐
 nearly ☐
 fairly well ☐
 exactly ☐

Q6 If you ticked NOT AT ALL in Question 5 the reason is: (tick all that apply)

- standard is out of date ☐
 standard is incomplete ☐
 standard is too academic ☐
 standard is too superficial ☐
 title is misleading ☐
 I made the wrong choice ☐
 other

Q7 Please assess the standard in the following categories, using the numbers:

- (1) unacceptable,
 (2) below average,
 (3) average,
 (4) above average,
 (5) exceptional,
 (6) not applicable

- timeliness
 quality of writing.....
 technical contents.....
 logic of arrangement of contents
 tables, charts, graphs, figures.....
 other

Q8 I read/use the: (tick one)

- French text only ☐
 English text only ☐
 both English and French texts ☐

Q9 Please share any comment on any aspect of the IEC that you would like us to know:

.....



ISBN 2-8318-8217-6



ICS 31.060.30
